

Yinghui He

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EDUCATION

- Princeton University** *September 2024 – June 2029 (Anticipated)*
Ph.D. Candidate in Computer Science. Advised by Sanjeev Arora. Princeton, NJ
- University of Michigan** *September 2022 – May 2024*
B.S.E. in Computer Science & B.S. (Minor) in Mathematics. (GPA: 3.98 / 4.00) Ann Arbor, MI
- Shanghai Jiao Tong University** *September 2022 – August 2024*
B.E. in Electrical and Computer Engineering. (GPA: 3.93 / 4.00; Ranking: 1 / 221) Shanghai, China

PROJECTS & PUBLICATIONS

Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision *ICML 2026 RLxF Workshop (Oral); AI4Math Workshop (Honorable Mention)*
Yinghui He, , Simran Kaur, Adithya Bhaskar, Yongjin Yang, Jiarui Liu, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, and Sanjeev Arora. *Princeton University*

- Proposed SD-Zero, an on-policy self-distillation method that converts binary rewards into dense token supervision.
- Conducted experiments on Qwen3 and Olmo3 models across math and code benchmarks and brought 10% gains.

Skill-Targeted Adaptive Training *ICLR 2026*
Yinghui He*, Abhishek Panigrahi*, Yong Lin, and Sanjeev Arora. *Princeton University*

- Proposed STAT, an adaptive training strategy that constructs training data targeted at models' missing skills.
- Solely implemented data creation pipeline; conducted SFT and RL training on Llama and Qwen models.

AdaptMI: Adaptive Skill-based In-context Math Instruction for Small Language Models *COLM 2025*
Yinghui He, Abhishek Panigrahi, Yong Lin, and Sanjeev Arora. *Princeton University*

- Introduced AdaptMI, an adaptive method that boosts in-context performance by 6% through skill-based examples.
- Solely conducted experiments on Llama and Qwen models (1B–7B), and large-scale skill labeling with GPT-4o.

Hi-ToM: A Benchmark for Evaluating Higher-Order Theory of Mind Reasoning in LLMs *EMNLP 2023*
Yinghui He, Yufan Wu, Yilin Jia, Rada Mihalcea, Yulong Chen, and Naihao Deng. *University of Michigan*

- Released the Hi-ToM benchmark that tests the higher-order theory of mind abilities in LLMs ([cited by 100+](#)).
- Independently implemented an automated data generator; coordinated human annotation on 600+ samples.

EmoAgent: Assessing and Safeguarding Human-AI Interaction for Mental Health Safety *EMNLP 2025*
Jiahao Qiu*, Yinghui He*, Xinzhe Juan, Yimin Wang, Yuhan Liu, Zixin Yao, Yue Wu, Xun Jiang, Ling Yang, and Mengdi Wang. *Princeton University*

- Built EmoAgent, a multi-agent AI framework to mitigate mental health risks by over 50% in human–AI interactions.
- Co-led experimental design and result analysis; coordinated the EmoAgent team.

Making Complex Reasoning Student-Friendly: A Hybrid Distillation Framework *Under Submission (2026)*
Yongjin Yang, Yinghui He, Jiarui Liu, and Zhijing Jin. *Princeton University*

- Proposed SFD, a hybrid distillation method that switches from teacher to student using entropy/NLL-based gating.
- Implemented dynamic student rewrite to reduce distribution shift, applying to 6 reasoning benchmarks.

LongProc: Benchmarking Long-Context Language Models on Long Procedural Generation *COLM 2025*
Xi Ye, Fangcong Yin*, Yinghui He*, Joie Zhang*, Howard Yen*, Tianyu Gao, Greg Durrett, and Danqi Chen. *Princeton University*

- Introduced LongProc, a benchmark that tests both long-form generation and highly dispersed information.
- Independently designed and generated 2k examples for the *Theory-of-Mind Tracking* subset, and evaluated 17 LLMs.

INTERNSHIP

- **NVIDIA** May 2026 – August 2026
Research Scientist Intern (Incoming) Santa Clara, CA
 - Expected to work on the Reinforcement Learning as a Pretraining Objective project, developing and evaluating methods to integrate RL signals into large-scale pretraining.
- **VMware** July 2022 – October 2022
Research Intern Shanghai, China
 - Studied algorithms and applications of federated analytics; proposed optimization methods on federated analytics.

SCHOLARSHIPS

- **Chinese National Scholarship**, Ministry of Education of the People's Republic of China ¥8000 (\$1170)
 - Highest scholarship in China awarded to Top 1% college students.
- **Fan Xuji Scholarship**, Shanghai Jiao Tong University ¥20000 (\$2940)
 - Highest SJTU scholarship awarded to 18 all-round developed students.
- **SJTU Undergraduate Excellence Scholarship**, Shanghai Jiao Tong University ¥1000 (\$150)
 - Awarded to top-ranking (10%) students.
- **John Wu and Jane Sun Excellence Scholarship**, UM-SJTU Joint Institute ¥75000 (\$11000)
 - Highest scholarship in UM-SJTU Joint Institute awarded to 10 all-round developed students.
- **The Jackson and Muriel Lum Scholarship**, UM-SJTU Joint Institute \$14000
 - Awarded to 5 excellent students in the dual-degree program.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, C#, Java, JavaScript, HTML/CSS, SQL
- **Model Training:** Multi-node & distributed training, Verl, TRL (Transformers Reinforcement Learning), LoRA, DeepSpeed, PyTorch, TensorFlow, scikit-learn, NumPy, pandas
- **Model Inference:** vLLM, Hugging Face Transformers, Accelerate, quantization & efficient serving
- **Agentic Frameworks:** LangChain, AutoGen, Hugging Face Hub, APIs
- **Data Science & Mathematical Tools:** pandas, NumPy, Matplotlib, Seaborn, SciPy, Jupyter, Colab, MATLAB, R, Mathematica, SymPy

HONORS AND AWARDS

- **Dean's List** 2023, 2024
Outstanding GPA, University of Michigan
- **Mathematical Contest in Modelling** Finalist Award (2%), May 2022
Consortium for Mathematics and its Applications
- **University Physics Competition** Bronze Medal, December 2021
UPC Committee
- **Media University-Enterprise Global Innovation Competition** First Place, November 2021
Media Shanghai
- **United Nations Global Youth Leadership Academy Summer Learning Program** Outstanding Participant, August 2021
GYLA

TEACHING & ADVISING

- **Teaching Assistant** September 2021 – August 2022
Physics II (60 students), Chemistry I (150 students) Shanghai Jiao Tong University
- **Student Advisor** September 2021 – August 2022
UM-SJTU Joint Institute Advising Center Shanghai Jiao Tong University
- **Writing Consultant** September 2021 – August 2023
UM-SJTU Joint Institute Writing Center Shanghai Jiao Tong University

EXTRACURRICULAR ACTIVITIES

- **Society of Women Engineers (SWE)**, Member
- **Princeton Koko Pops Dance Company**, Member
- **UMich Heartbeat A Cappella Group**, Song Arranger, Soprano
- **UMich Moli Dance Group**, Member